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Ching Him Leung

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Professional Position

May 2024 - present
Hall C Postdoctoral Fellow, *Thomas Jefferson National Accelerator Facility*
- Contribute to the commissioning and performance validation of GEM-based tracking detectors for high-rate experiments.
- Perform detector setup, calibration, and data-quality checks for the LAD experiment in Hall C.
- Conduct feasibility and sensitivity studies for proposed measurements (e.g., SoLID) using simulation-based evaluation of experimental configurations.
- Lead testing of micropattern gaseous detectors under high-rate operating conditions.

Education

Aug 2018 - May 2024
Ph.D. in Physics, *University of Illinois Urbana-Champaign*
Advisor: Prof. Jen-Chieh Peng
Thesis title: Probing Parton Distributions in Proton using Drell-Yan and Charmonium Production with 120 GeV Proton Beam at Fermilab

July 2014 - June 2018
Bachelor of Science, Physics, *The Chinese University of Hong Kong*

Other Research Experience

June 2019 - May 2024
Research Assistant, *University of Illinois, Urbana-Champaign*
FNAL-E906/SeaQuest
The aim of the experiment is to measure the light sea-quark asymmetry in proton using the Drell-Yan process.
- Analyzed the charmonium production data to extract the $p + d$ and $p + p$ differential cross sections, and the $(p + d)/2(p + p)$ charmonium cross-section ratio as an independent method to probe the $\bar{d}/\bar{u}$ asymmetry in the proton.
- Extracted the $(p + d)/2(p + p)$ Drell-Yan cross-section ratio from the entire SeaQuest datasets to determine the $\bar{d}/\bar{u}$ light sea-quark flavor asymmetry in proton for $0.1 < x < 0.45$.
- Developed an analysis procedure to ensure the statistical uncertainty from the simulation is properly included to extract the charmonium and Drell-Yan yields.
- Modified the Monte Carlo generator to ensure proper kinematic constraints are applied.
- Prepared internal analysis notes and presented progress reports to collaboration meetings.

FNAL-E1039/SpinQuest
The follow-up experiment of SeaQuest, which aims to measure the Sivers asymmetry of the sea-quark using the Drell-Yan process with a polarized target.
- Migrated the data acquisition system from SL6 to SL7.
- Managed and maintained the SpinQuest DAQ and computing systems.

May -Aug 2016
Summer Student, *University of Illinois, Urbana-Champaign*
- Analyzed transverse momentum distribution from existing Drell-Yan production data under the supervision of Prof. Jen-Chieh Peng.
- Performed fixed order QCD calculation for Drell-Yan process and compared with data.

Teaching Experience

Aug 2018 - May 2019 **Teaching Assistant**, *University of Illinois, Urbana-Champaign*
- Modern Experimental Physics (Spring 2022)
- Classical Mechanics II (Spring 2019)
- Atomic Phys & Quantum Theory (Fall 2018)

July - Aug 2015 **Junior Research Assistant**, *Faculty of Education, The Chinese University of Hong Kong*
Assisted instructors in conducting enrichment course in the summer Program for the Gifted and Talented 2015.

Honors and Awards

2022 **Felix T. Adler Award**, *University of Illinois, Urbana-Champaign*
Awarded for outstanding research performance by a graduate student in nuclear physics.

2020 **Maurice Goldhaber Research Scholar Award in Nuclear Physics**, *University of Illinois, Urbana-Champaign*
Awarded to graduate students demonstrating exceptional achievement and sustained excellence in nuclear physics research.

Publications

- K. Nagai et al. (FNAL E906/SeaQuest Collaboration), *Measurements of angular distributions of Drell-Yan dimuons in $p + p$ and $p + d$ interactions at 120 GeV/c*, (Jan. 23, 2026) arXiv:`2601.16410 [nucl-ex]`, preprint
- C. H. Leung, J. Dove, K. Nagai, et al. (FNAL E906/SeaQuest Collaboration), "Final SeaQuest Results on the Flavor Asymmetry of the Proton Light-Quark Sea with Proton-Induced Drell-Yan Process", Phys. Rev. Lett. **136**, 171901 (2026), arXiv:`2512.17564 [hep-ex]`
- C. H. Leung et al. (FNAL E906/SeaQuest Collaboration), "Measurement of J/ψ and $\psi(2S)$ production in $p + p$ and $p + d$ interactions at 120 GeV", Phys. Lett. B **858**, 139032 (2024), arXiv:`2406.11459 [hep-ex]`
- J. Dove et al. (FNAL E906/SeaQuest Collaboration), "Measurement of flavor asymmetry of light-quark sea in the proton with Drell-Yan dimuon production in $p + p$ and $p + d$ collisions at 120 GeV", Phys. Rev. C **108**, 035202 (2023), arXiv:`2212.12160 [hep-ph]`
- C. H. Leung, "Measurement of Charmonium Production in $p + p$ and $p + d$ Interactions in the Fermilab SeaQuest Experiment", in 29th International Workshop on Deep-Inelastic Scattering and Related Subjects (July 2022), arXiv:`2207.05640 [hep-ex]`
- J. Dove et al. (FNAL E906/SeaQuest Collaboration), "The asymmetry of antimatter in the proton", Nature **590**, [Publisher's correction: Nature 604, E26 (2022)], 561 (2021), arXiv:`2103.04024 [hep-ph]`
- A. Chen, J. Peng, C. Leung, et al. (FNAL E906/SeaQuest Collaboration), "Probing nucleon's spin structures with polarized Drell-Yan in the Fermilab SpinQuest experiment", PoS **SPIN2018**, 164 (2019), arXiv:`1901.09994 [nucl-ex]`
- W. Gong et al., "Stable Intrinsic Long Range Antiferromagnetic Coupling in Dilutely V Doped Chalcopyrite", Chinese Phys. Lett. **37**, 027501 (2020), arXiv:`1902.05281 [cond-mat.mtrl-sci]`

Conference Presentations

Jan 2025 **Probing the proton structure with Drell-Yan process and charmonium production at the SeaQuest experiment**
Invited talk at Cold Nuclear Matter Effects: from the LHC to the EIC Workshop.

March 2023 **Drell-Yan and charmonium production in $p + p$ and $p + d$ interaction at $120\,\mathrm{GeV}$ from the SeaQuest experiment**
Contributed talk at DIS2023: XXX International Workshop on Deep-Inelastic Scattering and Related Subjects.

Oct 2022 **Drell-Yan and charmonium production in $p + p$ and $p + d$ interactions at $120\,\mathrm{GeV}$ from the SeaQuest experiment**
Contributed talk at 2022 Fall Meeting of the APS Division of Nuclear Physics.

May 2022 **Measurement of charmonium production in $p + p$ and $p + d$ interaction in the Fermilab SeaQuest experiment**
Contributed talk at DIS2022: XXIX International Workshop on Deep-Inelastic Scattering and Related Subjects.

Oct 2021 **Measurement of J/ψ and ψ' production in $p + p$ and $p + d$ interaction with $120\,\mathrm{GeV}$ proton beam in the Fermilab SeaQuest experiment**
Contributed talk at 2021 Fall Meeting of the APS Division of Nuclear Physics.

Oct 2020 **Measurement of $p + d/p + p$ J/ψ cross section ratio with $120\,\mathrm{GeV}$ proton beam in the SeaQuest experiment**
Contributed talk at 2020 Fall Meeting of the APS Division of Nuclear Physics.

Technical Skills

Programming C/C++, Python, SQL

Tools Git, Apptainer/Singularity

HEP Library ROOT

OS Linux